Lack of efficacy of a reduced microparticle diet in a multi-centred trial of patients with active Crohn's disease.

BACKGROUND AND AIMS: Dietary microparticles, which are bacteria-sized and non-biological, found in the modern Western diet, have been implicated in both the aetiology and pathogenesis of Crohn's disease. Following on from the findings of a previous pilot study, we aimed to confirm whether a reduction in the amount of dietary microparticles facilitates induction of remission in patients with active Crohn's disease, in a single-blind, randomized, multi-centre, placebo controlled trial. METHODS: Eighty-three patients with active Crohn's disease were randomly allocated in a 2 x 2 factorial design to a diet low or normal in microparticles and/or calcium for 16 weeks. All patients received a reducing dose of prednisolone for 6 weeks. Outcome measures were Crohn's disease activity index, Van Hees index, quality of life and a series of objective measures of inflammation including erythrocyte sedimentation rate, C-reactive protein, intestinal permeability and faecal calprotectin. After 16 weeks patients returned to their normal diet and were followed up for a further 36 weeks. RESULTS: Dietary manipulation provided no added effect to corticosteroid treatment on any of the outcome measures during the dietary trial (16 weeks) or follow-up (to 1 year); e.g., for logistic regression of Crohn's disease activity index based rates of remission (P=0.1) and clinical response (P=0.8), in normal versus low microparticle groups. CONCLUSIONS: Our adequately powered and carefully controlled dietary trial found no evidence that reducing microparticle intake aids remission in active Crohn's disease.